BRD_Agent_V2 – Run It Locally

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You can run the whole flow on your Windows laptop in about 15 minutes, using Vertex AI for the model and local folders instead of GCS and Firestore.

1. Before you start

You need four things in place. Check each in PowerShell.

Need	Check with	Expected
Python 3.11 or newer	`python --version`	3.11+ (3.14 works)
Google Cloud CLI	`gcloud --version`	Prints a version
Access to project heypm-511616	`gcloud projects describe heypm-511616`	Project details, no error
Vertex AI User role on that project	Console > IAM	Your account listed with Vertex AI User

The code lives in `C:\Users\nikhi\Desktop\NIKHIL\GoogleAI\BRD_Agent_V2`. All commands below assume that path.

2. One-time setup

1. Open PowerShell in the parent folder and create a virtual environment (skip if you already use one for ADK):

```
cd C:\Users\nikhi\Desktop\NIKHIL\GoogleAI
python -m venv .venv
.venv\Scripts\activate
```

2. Install the dependencies:

```
pip install -r BRD_Agent_V2\requirements.txt
```

3. Sign in to Google Cloud so the code can call Vertex AI with your identity:

```
gcloud auth application-default login
gcloud auth application-default set-quota-project heypm-511616
gcloud services enable aiplatform.googleapis.com --project heypm-511616
```

4. Check `BRD_Agent_V2\.env`. It is already set up for local runs. These lines matter now:

Setting	Value for local runs	Why
GOOGLE_GENAI_USE_VERTEXAI	TRUE	Use Vertex AI, not an API key
GOOGLE_CLOUD_PROJECT	heypm-511616	Your project
GOOGLE_CLOUD_LOCATION	us-central1	Try `global` if the model is not found
STORAGE_BACKEND	local	Folders on disk instead of GCS and Firestore
DRY_RUN	true	Confirm runs end to end, but no email is sent and JIRA is not touched

With `STORAGE_BACKEND=local`, everything lives under `BRD_Agent_V2\local_data`:

Folder	Holds
transcripts\	Meeting transcripts, named `<meeting_id>.txt` (or .vtt, .json, .md)
templates\	Your BRD templates (.docx, .md, .txt)
meetings\	One JSON file per meeting: the Firestore record, stored locally
outputs\	Generated BRDs (.docx and .md), one folder per meeting

3. Run the full flow with the sample meeting

A sample transcript is already in place, with meeting id `MTG-2026-10-15-INVOICE-ALERTS`. Run these from inside the agent folder:

```
cd C:\Users\nikhi\Desktop\NIKHIL\GoogleAI\BRD_Agent_V2
```

Step 1: Process the meeting (what happens when the PM opens it). Takes 30 to 90 seconds.

```
python run.py process MTG-2026-10-15-INVOICE-ALERTS
```

You should see the record printed: `summary`, `MOM`, `jira_actions_requirend: true`, `jira_action_description` with actions A1, A2..., `relevant_jira` with FINAUTO-1098 and FINAUTO-1142, and `bdr_generated: false`.

Step 2: Look at the record (what the UI reads). Add `--full` to also see `agent_data`.

```
python run.py show MTG-2026-10-15-INVOICE-ALERTS
```

Step 3: Chat about the meeting. Ask anything, for example "What is the SLA for discount invoices?" or "Who owns the holiday calendar?". Press Enter on an empty line to quit.

```
python run.py chat MTG-2026-10-15-INVOICE-ALERTS
```

If you type "create the JIRA stories" in the chat, it shows the JIRA plan and asks you to confirm, the same way the UI's Confirm button would.

Step 4: Confirm the actions (the PM clicks Confirm). Skip the actions that touch tickets which don't exist in your JIRA.

```
python run.py confirm MTG-2026-10-15-INVOICE-ALERTS --skip A3,A4
```

In dry-run mode, created issues get placeholder keys such as `FINAUTO-NEW1`, and the MOM shows as `dry_run` with the recipients it would go to. `user_confirmation_provided` becomes true.

Step 5: Generate a BRD (only when the PM asks for one).

```
python run.py templates
python run.py brd MTG-2026-10-15-INVOICE-ALERTS
```

The BRD is written to `local_data\outputs\MTG-2026-10-15-INVOICE-ALERTS\` as .docx and .md, and `bdr_generated` becomes true. Use `--template "<name>"` to pick a specific template.

Start over: delete `local_data\meetings\MTG-2026-10-15-INVOICE-ALERTS.json`. To re-process a meeting that was already confirmed without deleting it, add `--force` to the process command.

4. Use your own meetings and templates

A new meeting: save its transcript into `local_data\transcripts\` with the meeting id as the file name, then process it.

```
copy C:\path\to\export.vtt local_data\transcripts\MTG-2026-10-20-SPRINT-
PLANNING.vtt
python run.py process MTG-2026-10-20-SPRINT-PLANNING
```

Accepted formats: Teams-style text (`Name m:ss` then the line spoken), WebVTT exports from Teams, Meet or Zoom, and JSON (a list of `{speaker, start, text}`). A transcript can also sit in a folder named after the meeting id.

People for the MOM email: add every attendee to `people.yaml` with name, email and optional JIRA account id. Anyone missing is reported as unresolved and gets no email or assignment. For a safe live test, point every email at yourself.

Your own BRD template: put a .docx, .md or .txt file in `local_data\templates\`.

- Word templates: use Heading 1/2/3 styles for sections; guidance paragraphs and table header rows are kept.
- Name a file `default.docx` (or `default.md`) to make it the template used when the PM doesn't pick one.
- Without any template, the bundled 20-section `templates\default_brd_template.md` is used.

```
python run.py templates
python run.py brd MTG-2026-10-20-SPRINT-PLANNING --template "Finance BRD"
```

5. Try it in the ADK dev UI

The dev UI gives you a chat window with the same agent, and shows confirmations as dialogs.

1. Start it from the parent folder, not from inside BRD_Agent_V2:

```
cd C:\Users\nikhi\Desktop\NIKHIL\GoogleAI
adk web
```

2. Open http://localhost:8000 and pick **BRD_Agent_V2** in the agent dropdown.
3. Click **New Session** and type: `open meeting MTG-2026-10-15-INVOICE-ALERTS`.

4. Ask questions, or say "process the meeting", "create the JIRA stories" (approve in the dialog) or "generate the BRD".

Stop it with Ctrl+C, and restart it after changing `.env` or any code.

6. Going live, one step at a time

Change one thing, run the flow again, then move to the next.

Real JIRA

1. Create an API token at id.atlassian.com > Security > API tokens.
2. In `.env`, set `JIRA_BASE_URL`, `JIRA_EMAIL`, `JIRA_API_TOKEN` and `JIRA_PROJECT_KEY` (the prefix on your ticket keys).
3. Set `DRY_RUN=false`, then confirm a meeting and skip any action that touches tickets missing from your JIRA. If an existing ticket in the plan can't be found, nothing is changed and the error says which one.

Real email (Gmail)

1. Turn on 2-Step Verification, then create an app password at myaccount.google.com/apppasswords.
2. In `.env`: `SMTP_HOST=smtp.gmail.com`, `SMTP_PORT=587`, `SMTP_USERNAME` and `MAIL_FROM` = your Gmail address, `SMTP_PASSWORD` = the 16-character app password.

GCS and Firestore instead of local folders

```
gcloud services enable firestore.googleapis.com storage.googleapis.com --
project heypm-511616
gcloud firestore databases create --location=eur3 --project heypm-511616
gcloud storage buckets create gs://<your-bucket> --location=europe-west1 --
project heypm-511616
gcloud storage cp local_data\transcripts\MTG-2026-10-15-INVOICE-ALERTS.txt
gs://<your-bucket>/transcripts/
```

Then in `.env` set `STORAGE_BACKEND=gcp` and `TRANSCRIPT_BUCKET=<your-bucket>`, and run the same commands as in section 3. The record now appears in Firestore under `meetings/MTG-2026-10-15-INVOICE-ALERTS`, and BRDs go to `gs://<your-bucket>/brds/<meeting_id>/`. Skip the database-create command if your project already has a Firestore database.

Keep `.env` private once it holds tokens and passwords.

7. Troubleshooting

Symptom	Likely cause	Fix
`TranscriptNotFound`	File name doesn't match the meeting id	Name it `<meeting_id>.txt` (or .vtt/.json/.md) in `local_data\transcripts\`
403 / permission denied from Vertex	Missing role or API	Add Vertex AI User to your account; enable aiplatform.googleapis.com
404 model not found	Model not offered in the region	Set `GOOGLE_CLOUD_LOCATION=global` in `.env`
`AlreadyConfirmed` on process	Meeting already confirmed	Add `--force`, or delete its JSON in `local_data\meetings\`
`MeetingNotProcessed` on confirm	Confirm before process	Run `process` first
MOM `unresolved` names, nobody emailed	Attendees missing from people.yaml	Add them with emails
JIRA: "existing issues were not found"	Plan links to tickets that aren't in your JIRA	Re-run confirm with `--skip` for those refs
BRD warning "looks incomplete"	Generation cut short	Run `brd` again; send the warning if it repeats
BRD_Agent_V2 not in `adk web` dropdown	Started from the wrong folder	Run `adk web` from the GoogleAI folder
Changes to `.env` ignored	Process still running with old settings	Stop and restart `run.py` or `adk web`

Run the offline tests to check the install without using the model:

```
pip install pytest pytest-asyncio
pytest -q
```

All 8 tests should pass.